Lesson 2.6 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Median and Outliers · I can explain why the median is a better measure of center than the mean when a data set has an outlier.

S2: Comparing Distributions · I can compare two distributions using the median and the interquartile range together.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Median and Outliers

I can explain why the median is a better measure of center than the mean when a data set has an outlier. · 6.SP.B.5d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	Order: 3, 20, 21, 22, 23, 24, 25. a. 3 b. 22	Ask whether the question is about a central observation, a fair share, or a total; use the matching evidence.
2	The median, $5. The $40 outlier pulls the mean up to $12, more than 4 of the 5 prices.	Ask whether the question is about a central observation, a fair share, or a total; use the matching evidence.
3	B — more than the median	Ask the learner to show why B — more than the median is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Uses 'outlier means median' without considering the purpose or computing the summaries..

Response: Ask whether the question is about a central observation, a fair share, or a total; use the matching evidence.

Also check: Calling the outlier the typical value because it stands out.

Also check: Thinking a bigger outlier always makes the median bigger.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Compare mean and median of 2, 3, 4, 5, 6 with 2, 3, 4, 5, 26.

Teacher solution: First mean/median 4/4; second mean/median 8/4. Explain the different sensitivity to the extreme change.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: A: mean 6, median 6. B: mean 10, median 6. The median did not change; the mean increased by 4. For a central individual value, the median is more representative here, while the mean still summarizes total divided by count.

Misconception response anchor: Uses 'outlier means median' without considering the purpose or computing the summaries.. Use ask whether the question is about a central observation, a fair share, or a total; use the matching evidence.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: For the purpose of ___, I use ___ because the change in the extreme value ___.

Session 2 · Comparing Distributions

I can compare two distributions using the median and the interquartile range together. · 6.SP.B.5d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. Shop K b. same (both IQR 8)	Write one sentence about center and one about spread, then connect the recommendation to a stated priority.
2	8; 12 − 5 = 7	Write one sentence about center and one about spread, then connect the recommendation to a stated priority.
3	C — E is higher; F is more consistent	Ask the learner to show why C — E is higher; F is more consistent is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Chooses the lower median without considering variability or claims a guaranteed wait..

Response: Write one sentence about center and one about spread, then connect the recommendation to a stated priority.

Also check: Thinking the group with the higher median is also the more consistent one.

Also check: Comparing only medians and ignoring the IQRs.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A service has median 12 and IQR 3 minutes; another has median 9 and IQR 10. Explain one reason to choose each.

Teacher solution: First for consistency of the middle half; second for the lower typical time. State that both measures summarize distributions, not guarantees.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: A has the shorter typical wait by median, 3 minutes less. B has the narrower middle-half spread, by 6 minutes. Either choice can be justified by the stated priority; neither summary guarantees an individual wait.

Misconception response anchor: Chooses the lower median without considering variability or claims a guaranteed wait.. Use write one sentence about center and one about spread, then connect the recommendation to a stated priority.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: If the priority is ___, I recommend ___ because ___. The tradeoff is ___.